# Limitations and Ethics

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Large Language Models for Health and Social Science Research, June-July 2026

## From Lecture 1 to today

Two things you already have, that today leans on hard.

- **Lecture 1:** text is a lossy trace of a phenomenon; every downstream transformation obeys

$$I(S; T(O)) \leq I(S; O).$$

  Bigger, more capable models do not repeal this. They make the loss harder to see.

- **Lecture 3:** foundation models *homogenize* — one model's flaws propagate into every system built on it, and you carry real obligations (governance, documentation) when you adopt one.

### Today's question

What happens — ethically, and then epistemically — when a lossy, homogenized artefact is deployed at the scale of a search engine, a newsroom, an election, or a research pipeline?

# Lecture roadmap

1. Four familiar ethical concerns, each grounded in a real case
   - Environmental and financial cost
   - Training data: consent, ownership, whose text
   - Malicious and unethical application
   - Risky application inside research itself

   — **break** —

2. A sharper, epistemological version of concern (2)+(4): what is a language model actually doing when we treat its output as evidence about the world?
   - Synthetic respondents in social science
   - Semantic uncertainty: why the standard tools don't diagnose this

The second half draws on ongoing work, including my own — treat it as live research, not settled fact.

# Why case studies, not just principles

- Ethical AI critique is easy to nod along to in the abstract and easy to forget in practice.
- Each concern below is attached to a documented, named incident — with real financial, legal, or reputational consequences — precisely so it resists becoming abstract.
- Recurring question across all four: *who bears the cost, and who captures the benefit?* Keep this in mind through the whole section.

# Concern 1: environmental and financial cost

- Training a large model is not a one-off cost paid by its creator alone — energy, water, and rare-earth hardware demand are borne by the grids and communities hosting data centres.
- Bender et al. open "Stochastic Parrots" with exactly this point, before getting to language at all: *scale has a physical cost, and that cost is currently invisible in how we evaluate models.*
- This is not a solved problem from 2019 — it has scaled up with the models.

### The question to hold onto

Who pays this cost, and who captures the benefit of the resulting capability?

# Case study: Strubell et al. (2019)

**Energy and Policy Considerations for Deep Learning in NLP** — the paper Bender et al. cite for their environmental argument.

- Estimated the $CO_2$ emissions of training several NLP architectures of the era (Transformer, ELMo, BERT, GPT-2).
- Headline number: training a single large Transformer with neural architecture search emitted an estimated **626,000 lbs of $CO_2$** — roughly the lifetime emissions of **five cars** (including manufacture).
- Financial cost estimates ran into the hundreds of thousands of dollars for a single training run, *before* the many further runs needed for hyperparameter search.

**Why this matters beyond 2019:** models discussed in Lectures 2–3 are one to three orders of magnitude larger. The relative point — cost scales with capability, and is rarely priced into how "success" is reported — still holds.

# Concern 2: training data — consent, ownership, whose text

Recall Lecture 3's **Data Statements** discussion (Bender & Friedman): pretraining corpora almost never document *whose* language they contain.

- Large models are trained on web-scraped text, mostly without the authors' knowledge or consent.
- This raises two distinct questions that are often conflated:
  - **Legal:** does this infringe copyright? (unsettled, jurisdiction-dependent, actively litigated)
  - **Normative:** even where legal, is it fair to the people whose labour produced the text a commercial model profits from?
- Bender et al.'s critique of "uncurated" data (Lecture 3 callback) is precisely about not knowing — and not being able to consent to — what's inside the training set.

# Case study: *New York Times v. OpenAI and Microsoft*

Filed December 2023; still active as of this lecture.

- The Times alleges GPT models were trained on millions of its articles without permission, and can reproduce substantial portions verbatim on request.
- OpenAI's defence: training is **transformative fair use**; the Times allegedly had to manipulate prompts extensively to elicit verbatim reproduction.
- April 2025: Judge Stein allowed the core copyright claims to proceed to trial — the case is far from resolved.
- **Complication:** in 2025, courts in two *separate* cases (*Bartz v. Anthropic*, *Kadrey v. Meta*) found that training itself is highly transformative and can qualify as fair use — so the law is actively unsettled, not simply trending one way.

**Side effect worth flagging:** discovery in this case has forced OpenAI to preserve millions of user chat logs, creating a live tension between copyright litigation and *user* privacy — an unexpected second-order harm.

# Concern 3: malicious and unethical application

- Distinct from training-data harms: here the model works *as intended*, but is deployed to deceive, suppress, or manipulate.
- Generative capability (voice cloning, image/video synthesis, fluent text) lowers the cost of producing convincing disinformation to near zero.
- This is squarely inside Weidinger et al.'s "malicious use" risk category, which we'll return to after the break.

# Case study: the 2024 New Hampshire "Biden" robocall

Two days before the January 2024 NH Democratic primary, thousands of voters received a robocall in an AI-cloned voice of President Biden, telling them not to vote in the primary and to "save" their vote for November.

- Created by political consultant Steve Kramer for $150, using commodity AI voice-cloning tools.
- Caller ID was spoofed to appear as a local political operative — a deliberate deception layered on top of the deepfake itself.
- **Consequences:** FCC proposed a $6 million fine against Kramer; the telecom carrier Lingo Telecom paid a $1 million penalty for inadequate caller-verification; the FCC subsequently banned AI-voice robocalls outright.
- **Complication:** a New Hampshire jury *acquitted* Kramer of the criminal voter-suppression and impersonation charges in July 2025. The FCC fine and civil consequences remain.

**Discussion point:** the harm here required no model failure at all — fidelity was the entire attack.

# Concern 4: risky application inside research itself

The cluster most directly relevant to this room.

- Concerns 1–3 are about the world outside the lab. This one is about what happens when researchers themselves over-trust model output.
- The core risk: fluent, confident, well-formatted text is easy to mistake for verified content — especially in domains (science, medicine) where authoritative register is itself a strong credibility signal.
- Recall Lecture 1: NLP systems compress *form*, not truth. A model optimized to sound like a scientific paper has no built-in mechanism ensuring the claims inside are correct.

# Case study: Meta's Galactica (November 2022)

Marketed explicitly as a model that could "store, combine, and reason about scientific knowledge" — summarizing papers, writing literature reviews, generating citations.

- Trained on 48 million papers, textbooks, and scientific web text.
- Public demo launched 15 November 2022; **pulled within 48 hours**.
- Scientists immediately found it generating fluent, citation-studded, entirely **fabricated** content: fake papers attributed to real authors, false physical constants, confidently wrong claims in specialists' own fields.
- Michael Black (Max Planck Institute): *"In all cases, it was wrong or biased but sounded right and authoritative. I think it's dangerous."*

**Why this belongs in a Bender et al. discussion:** this is the "illusion of understanding" critique made concrete — a model with no mechanism for truth, deployed precisely where fluency is mistaken for expertise.

# Case study: the paper-mill / “AI slop” crisis, 2025–26

Galactica was a single dramatic failure, pulled in 48 hours. This is the slower, larger version of the same problem, unfolding across the literature right now.

- **Organization Science** (a top management journal) directly measured this: submissions up **42%** since ChatGPT’s release — roughly double the bump seen during COVID. Human-only submissions actually *declined*.
- By early 2026, a **majority** of submitted manuscripts show detectable AI involvement; the fastest-growing category is papers scoring 70%+ AI-generated.
- **Paper mills** — commercial operations selling ready-made manuscripts and guaranteed authorship slots — now use LLMs to mass-produce fabricated results, plausible-looking figures, and entire fake studies, concentrated in high-stakes fields like cancer research.
- Screening work on the PubMed corpus (17–20 million papers) is now needed just to *estimate* the scale of the problem.

# Why this one is different from Galactica

- Galactica was a *product* that failed publicly and was withdrawn. The slop crisis is a *practice* — ordinary researchers and paper mills alike, using ordinary tools, at industrial scale, with no single moment of correction.
- There is no clean line between "a researcher used an LLM to help draft a discussion section" and "a paper mill used an LLM to fabricate an entire study." Detection tools (Section 7 territory, Lecture 3) increasingly can't cleanly separate these either.
- **This is not a hypothetical for this room.** Every methods-section documentation habit from Lecture 3 (model, version, prompts, validation) exists precisely to keep your own work on the right side of this line, visibly.

# Where this leaves us

- Four concerns, four documented harms — none hypothetical, all with real financial, legal, or scientific consequences.
- Notice the common structure: in every case, **a genuine capability** (fluent generation, voice synthesis, broad text coverage) produces harm *via the specific way it was deployed or trusted* — not via a bug that better engineering straightforwardly fixes.
- Case 4 (Galactica) is the pivot point for the rest of today: it is a research-methods failure, not just a product failure.

### Epistemic Problem

If confident, fluent output can be mistaken for verified scientific content, what happens when a discipline's actual *methodology* starts asking language models to stand in for something else entirely — a human research participant? That's where we're going next.

10 min break

# From “is this harmful” to “what is this, epistemically”

Part 1 asked: does this output cause harm — financial, legal, scientific?

- That question can be answered by pointing at consequences.
- The next question is harder, and sits underneath many of Part 1’s cases: **when a model produces text, what — if anything — does that text tell us about the world it was trained to describe?**
- We’ll work through this with one live methodological debate in social science: using LLMs *as* survey respondents.

Recall Lecture 1

$S \to O \to T(O)$, and $I(S; T(O)) \leq I(S; O)$. Keep this exact structure in mind — we are about to see it appear again, in a place you might not expect.

# The promise: “silicon sampling”

Argyle et al. (2023) propose conditioning an LLM on a real respondent’s sociodemographic backstory, and asking it to predict that respondent’s answer to a held-out survey question.

- They term the result **algorithmic fidelity**: the model doesn’t just match marginal response rates, it reproduces fine-grained *correlations* between attitudes, conditional on demographic backstory.
- The appeal for social science is obvious — human samples are small and expensive; a well-conditioned LLM promises statistical power for a fraction of the cost.
- This is not a fringe idea: it has been extended to economics (“homo silicus,” Horton 2023), and **commercialized** directly — companies including Synthetic Users and VeraSight now sell LLM-generated “synthetic panels” to market researchers.

## The empirical cracks

If algorithmic fidelity held generally, this would mostly be an engineering problem. It doesn't.

- **Bisbee et al. (2024):** silicon-sampled GPT-3.5 responses inflate estimates of partisan and racial affective polarization by a factor of roughly **seven** relative to human benchmarks — attributed to stereotype amplification under persona conditioning.
- Performance gaps are worst precisely where stakes are highest: socially sensitive topics ("harmlessness bias" toward socially desirable answers), and non-Western/non-English contexts, tracking training-corpus imbalance.
- A synthesis of 285 silicon-vs-human comparisons (Sarstedt et al., 2024) concludes LLMs reproduce *surface-level* patterns but not deeper behavioural regularities.

**The methodological response:** rather than abandon the approach, researchers proposed ways to *correct* for this — which is where the real epistemological question hides.

# Two proposed fixes

1. **Check for exchangeability:** compare synthetic and human responses on some metric (correlation, distributional overlap); if they look similar enough, treat them as interchangeable and pool them.
2. **Debias via a gold-standard human subsample** (Broska et al., 2025, applying **Prediction-Powered Inference**, Angelopoulos et al. 2023): use a smaller human sample to correct a larger LLM-generated sample. If LLM bias is negligible, you gain efficiency; if not, estimates default toward the human-only result.

## Why fix (2) looks safe

It's framed as *always valid*: worst case, you fall back to the human sample. This is the claim we're going to interrogate.

**Both approaches share a hidden dependency:** they require treating LLM outputs as **i.i.d. draws** from (something close to) the same process that generates human responses. That assumption is where we go next.

# What i.i.d. actually buys you

Independent and Identically Distributed — the load-bearing assumption of frequentist inference (CLT, MLE, standard errors, all of it).

**Independence.** For events $A_1, \ldots, A_k$:

$$P\left(\bigcap_{j=1}^{k} A_j\right) = \prod_{j=1}^{k} P(A_j).$$

Beyond the algebra: each observation carries **non-redundant** information about the underlying process. More observations $\Rightarrow$ more knowledge.

**Identically distributed.**

$$P(A_1) = P(A_2) = \cdots = P(A_k).$$

Repeated observations are draws from the *same* underlying process — so it's licit to pool them.

**Both assumptions have to hold jointly** for pooling synthetic and human data to mean what standard inference needs it to mean.

# Reframing this as Lecture 1's DPI

Put the LLM pipeline into the $S \to O \to T(O)$ structure directly.

- $S$: the human social process — the actual attitudes, beliefs, and behaviours we want to learn about.
- $O$: the training corpus — one enormous, but ultimately **single**, historical trace of $S$.
- $T(O) = \theta$: the trained model's fixed parameters — a compressed, lossy summary of that one trace.

**The crucial fact:** $\theta$ is estimated *once*, at training time, and then **frozen**. Every subsequent query — every "synthetic respondent" you generate — is a query to this one fixed object, not a fresh interaction with $S$.

### The question this raises

If $\theta$ is fixed, what exactly are you sampling when you generate $J$ "different" synthetic respondents?

# Independence fails: there is only one draw

Treating $J$ LLM generations as $J$ independent new experiments requires each to come from an independent new interaction with $S$. They don't.

$$P(Y_{1:J} \mid \theta, p_{1:J}) \neq \prod_{j=1}^{J} P(Y_j \mid \theta_j), \qquad \theta_j = \mathcal{L}(\theta, \mathcal{D}_j).$$

- The right-hand side is what independence *would* require: each $Y_j$ generated by parameters $\theta_j$ updated on its own independent data $\mathcal{D}_j$.
- At inference time, there is no $\mathcal{D}_j$. $\theta$ does not update. Prompts $p_{1:J}$ **condition** a fixed distribution; they do not **inform** it.
- So all $J$ outputs are downstream of the *same* single draw — the training step — dressed up as $J$ different draws by varying the prompt.

**This is the DPI, restated as a sampling failure:** conditioning a fixed, already-lossy $T(O)$ cannot manufacture new independent information about $S$, no matter how many times you query it.

# Identically distributed also fails — but subtly

Compared only to *each other*, LLM outputs genuinely are i.i.d.: same model, same representational space, conditional on fixed decoding settings. That part is fine.

**The failure is at the joint level.** Claiming LLM outputs and human responses are identically distributed *together* requires the model's representational space to fully cover the space of human responses. In general:

$$\mathrm{supp}(P_{\mathsf{LLM}}) \subsetneq \mathrm{supp}(P_{\mathsf{human}}).$$

- $P_{\mathsf{LLM}}$'s support is finite and fixed by one historical corpus. $P_{\mathsf{human}}$ is generated by an open, ongoing social process that keeps producing genuinely new modes of response.
- Even where observed LLM outputs *happen to match* observed human responses, matching realizations does not establish that the generating distributions coincide.

# Why you can't just check this empirically

The natural response: "fine — let's test exchangeability empirically, using correlation, effective sample size, coverage."

## The circularity

Every one of those diagnostics is *itself defined* with respect to a data-generating process in which repeated observations are i.i.d. To measure "how exchangeable" two sources are, you must already have assumed the sampling structure whose validity is in question.

- Interchangeability is not a hypothesis the data can settle on its own — it is a **constitutive assumption** about what counts as "repeating the same experiment."
- Without prior epistemic licence to treat $\theta$'s outputs as draws from $S$, an empirical alignment score cannot retroactively grant that licence.

**Consequence for PPI-style debiasing (Broska et al.):** the "always valid, worst case you fall back to human data" framing is less safe than it looks — the debiasing step itself inherits the same i.i.d. dependency it was meant to route around.

# What this does *not* say

To be precise about the claim, because it's easy to overstate.

- **Not claimed:** LLMs can never be used in social-science measurement pipelines.
- **Claimed:** using LLM output *as if it were* an i.i.d. sample from the human population requires an epistemic licence that current empirical-check and debiasing methods assume, but do not establish — and in the case of the i.i.d. assumption specifically, cannot establish by the data alone.
- Other uses of LLMs sit on different epistemic footing — e.g. **classification** of existing human-generated text (annotation, coding), which doesn't ask the model to generate new "respondents" at all, just to label data that already came from $S$. This is closer to the design-based/semi-supervised settings from Lecture 2.

**The real question, restated:** not "can this be deployed," but *what inferential claims is a given use actually licensed to support*.

# The deeper question underneath this

Notice what the whole argument turned on: not that LLM outputs are *low quality*, but that repeated queries to a fixed $\theta$ don't behave like repeated sampling from $S$ — no matter how varied the prompts, how fluent the output, how well individual answers seem to match human data.

### Bridge to the next section

This raises a question we've been dodging: when you *do* get variation across queries to the same model — different phrasings, different sampling temperature, different-but-related prompts — what *is* that variation telling you? Is it telling you anything about the world, or only about the model's internal geometry?

This is exactly the question semantic uncertainty research tries to formalize — and it's where we're headed next.

# What does variation across queries actually measure?

We ended Part 2A on a question: when an LLM's output varies across prompts, phrasings, or sampling runs, what is that variation telling you?

- The standard tool for this in current NLP research is **semantic entropy** (Kuhn et al. 2023): sample several generations for one input, cluster them by *meaning* (usually via bidirectional entailment / NLI), and compute entropy over the resulting clusters rather than over raw strings.
- Low semantic entropy $\Rightarrow$ the model consistently says "the same thing," even if the exact wording varies $\Rightarrow$ treated as a proxy for confidence/reliability.
- **This method needs a working notion of "same meaning."** It borrows one implicitly, without asking which one, or whether it's the right one for the question being asked.

## Where we're going

There isn't one settled formal theory of meaning. Two very different ones give two very different answers to what "semantic stability" should even mean.

# Two formalizations of meaning

**Montague: model-theoretic**

- Meaning = truth conditions
- A fixed correspondence between expressions and a model
- No sender, no receiver, no interaction

**Lewis / Skyrms: coordination-based**

- Meaning = a convention that solves a coordination problem
- Constituted by repeated interaction between agents
- No fixed model — meaning is an equilibrium property

**The claim we're building to:** Kuhn et al.'s semantic entropy is implicitly Montagovian. That's a substantive choice, not a neutral default — and it may be measuring the wrong kind of stability for questions about whether a model's language can actually coordinate with anyone.

# Montague semantics, formalized

A model $M = \langle D, I \rangle$: a domain of individuals $D$, and an interpretation function $I$ assigning denotations to non-logical constants.

- Every well-formed expression $\alpha$ has a denotation $\alpha^M$, computed **compositionally**: the meaning of a complex expression is a function of the meanings of its parts and their mode of combination.
- A sentence $S$ is **true in** $M$ if $S^M = 1$; extended to possible-worlds semantics, meaning becomes a function from worlds to truth values (an intension).

## What this presupposes

A fixed, agent-independent correspondence between language and a domain. Two utterances have "the same meaning" if they have the same truth conditions in $M$ — full stop. There is no place in this picture for a sender, a receiver, or a history of interaction between them.

# Lewis / Skyrms, formalized

A **signaling game** $G = \langle T, M, A, \sigma, \rho, u \rangle$:

- $T$: states of the world; $M$: messages; $A$: receiver actions
- $\sigma : T \to M$, the sender's strategy; $\rho : M \to A$, the receiver's strategy
- $u$: a shared (or partially shared) payoff, maximized when $\rho(\sigma(t))$ is the correct action for state $t$

### Signaling system equilibrium

A pair $(\sigma, \rho)$ is a **signaling system** if $\sigma$ is injective and $\rho(\sigma(t)) = a(t)$ for every $t \in T$: distinct states get distinct messages, and the receiver always recovers the right action.

**Meaning of message** $m$ is *defined by its role in this equilibrium*: $m$ "means" $t$ if $\sigma(t) = m$. There is no model $M$ fixed in advance — meaning is constituted by the strategy pair itself.

# Skyrms's extension: meaning without prior rationality

Lewis (1969) treats signaling systems as solutions rational agents could reach by explicit reasoning. Skyrms asks: do you need that much?

- Skyrms shows signaling systems emerge from **simple reinforcement dynamics** (e.g. Roth–Erev learning) over repeated play, with no prior rationality or common knowledge required — just repeated interaction and payoff-following.
- **Multiple signaling systems can be equally good equilibria.** There is no guarantee the population converges to a unique one, or that whichever one it reaches tracks anything "true" about $T$ beyond successful coordination.
- Consequence: meaning, on this view, is inherently **social, dynamical, and potentially non-unique** — not a fixed fact about a static model.

# This is Lecture 1's Alice and Bob, formalized

Recall the encoding chain from Lecture 1:
$M_{\text{mind}} \to M_{\text{speech}} \to M_{\text{text}} \to M'_{\text{text}} \to \widehat{M}_{\text{speech}} \to \widehat{M}_{\text{mind}}$, and the warning that communication is the movement of *encoded representations*, not meaning itself.

- A Lewis/Skyrms signaling game *is* a formal model of exactly this chain — sender, message, receiver, success condition — made game-theoretic and dynamic.
- Montague semantics, by contrast, has no sender or receiver in it at all. It asks what a sentence *means*, as a static fact, once you already have a model $M$.
- **Neither is wrong.** They are answers to different questions: "what are the truth conditions of this string" vs. "does this string let two agents successfully coordinate."

# Where semantic entropy sits, and the gap it leaves

Kuhn et al.'s clustering step — group generations by bidirectional entailment — is a **Montagovian** move: it asks whether two strings have the same truth conditions, computed by a single auxiliary model (an NLI classifier), with no second agent and no interaction.

## The gap

Low semantic entropy (Montague-stable) does **not** imply the output would let a second agent correctly recover the intended state $t$ in a signaling game (Lewis/Skyrms-stable). These are logically independent notions of "stability," testing different things.

- A model can produce paraphrase-consistent, truth-condition-stable text about a contested term, while that text *fails* to coordinate two different interpreters onto the same referent — because the term's meaning, in Lewis/Skyrms terms, was never a single stable convention to begin with.
- Existing semantic uncertainty methods have no way to detect this failure mode, because they never construct a receiver.

# The ontological fork the framework forces

Go back to the signaling game: $(\sigma, \rho)$ is a signaling system — an **equilibrium** that a population of interacting agents settles into. Two very different things could correspond to "an LLM" in this picture.

### Option A: the LLM is an agent

It occupies a role in the game — sender or receiver — with its own strategy, potentially its own goals, something we could hold to a standard of good or bad play.

### Option B: the LLM is the equilibrium

It *is* $(\sigma, \rho)$ itself — a crystallized, statistical summary of how a population of *human* senders and receivers already coordinated meaning, frozen at training time (recall: $\theta$ is fixed, Part 2A).

**This is not a rhetorical choice.** It changes where ethical responsibility can coherently live.

# Two very different research programmes follow

## If the LLM is an agent

- Ethics is imposed *on the model*: alignment, constitutional training, corrigibility, "teaching it values"
- Failures are the agent's failures — it was insufficiently honest, insufficiently careful
- Fixes look like: better training objectives, better RLHF, better guardrails

## If the LLM is an equilibrium

- The model has no more moral standing than a compressed dictionary of how *we* already spoke
- Failures are *our* failures — whose speech got included, who deployed it, how, on whom
- Fixes look like: Lecture 3's governance, documentation, and data-curation practices

**Notice:** Option A is where most public "AI ethics" discourse currently sits. Bender et al.'s entire argument, restated in this vocabulary, is a case *for Option B*.

# Reading today's cases through the fork

Every case from Part 1 looks different depending which option you pick.

- **The robocall:** under Option A, you might ask whether the *voice-cloning model* behaved wrongly. It didn't — it worked exactly as designed. The wrongdoing was Kramer's, a human decision about deployment. **Option B was always the right frame here**, and the FCC's fines and charges, correctly, targeted people, not the tool.
- **Galactica / paper mills:** under Option A, the model "hallucinated." Under Option B, a compressed equilibrium of scientific-sounding text was deployed somewhere it could be mistaken for verified knowledge — a curation and deployment failure, squarely on the people who built and shipped it that way.
- **Synthetic respondents (Part 2A):** the entire critique *is* an Option-B argument — $\theta$ is a frozen equilibrium of a past corpus, not a fresh agent capable of independently informing you about the present social world.